

Literature Survey: Self-Auditing Clinical Decision Support Agent

Agentic AI for Trustworthy Clinical Decision Support / Texas Health Informatics Research

Field	Details
Project	Self-Auditing Clinical Decision Support Agent
Positioning	Agentic AI for Trustworthy Clinical Decision Support / Texas Health Informatics Research
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Architecture Visual Representations

The following visual representations summarize the proposed self-auditing clinical AI pipeline and the verification logic used to audit claims before clinical decision-support output is presented.

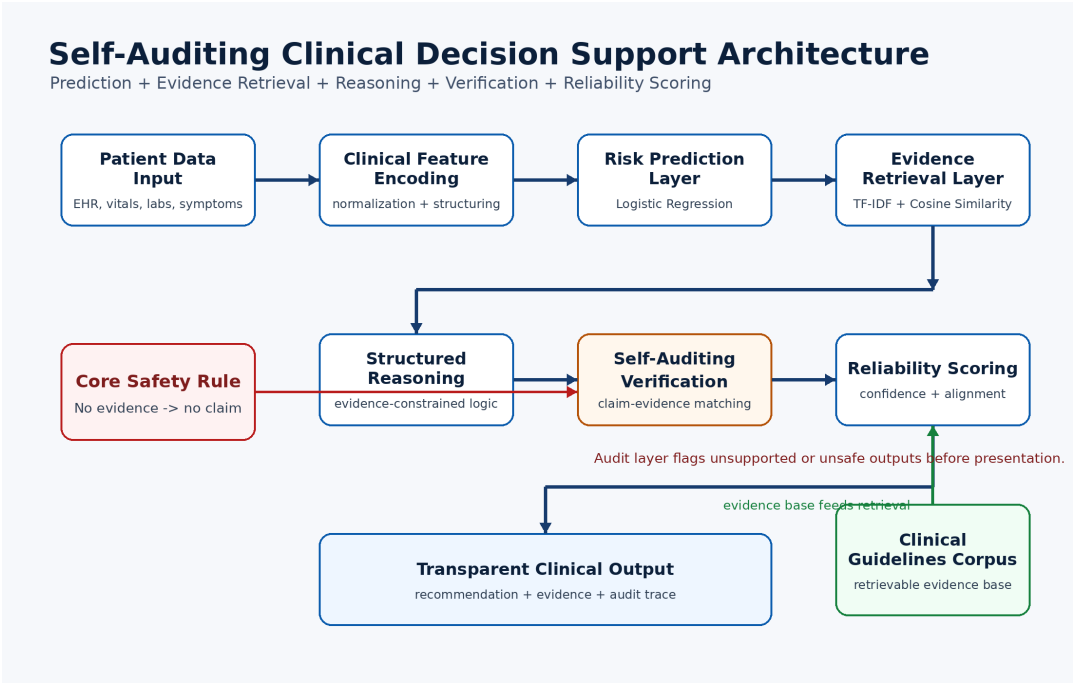


Figure 1. End-to-end architecture: patient data is transformed into transparent clinical decision-support output through prediction, retrieval, reasoning, verification, and reliability scoring.

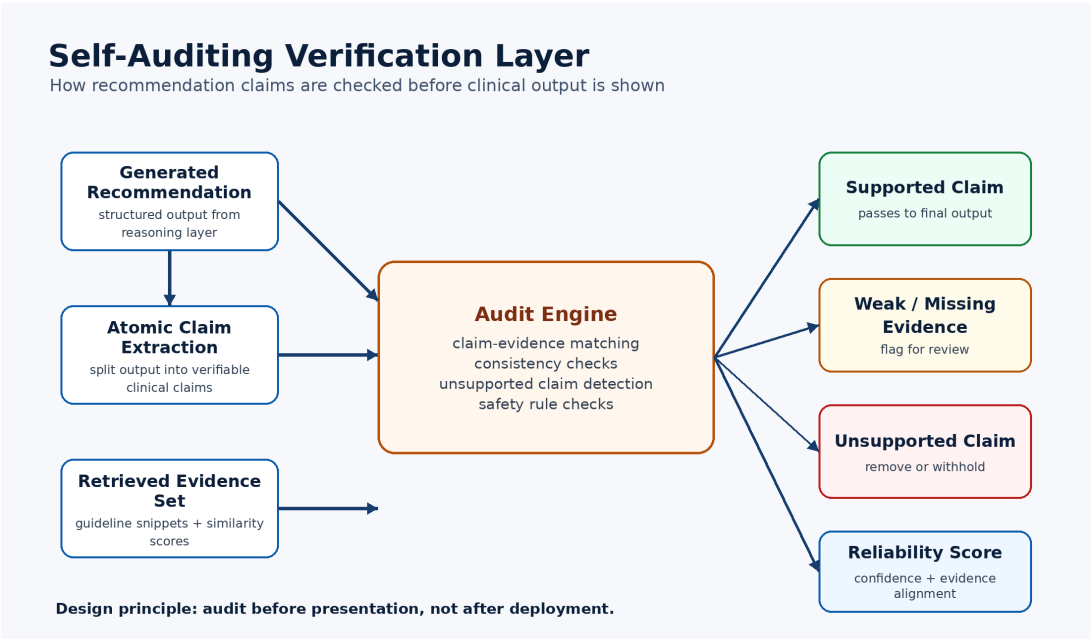


Figure 2. Self-auditing verification layer: generated claims are matched against retrieved evidence, flagged when unsupported, and summarized through reliability scoring.

Abstract

This literature survey examines research relevant to the design of a Self-Auditing Clinical Decision Support Agent—a modular healthcare AI architecture that integrates clinical risk prediction, evidence retrieval, structured reasoning, self-auditing verification, and reliability scoring. The survey is organized around a central research question: How can healthcare AI move from prediction-only systems to verifiable, evidence-grounded reasoning systems? Drawing on systematic reviews, consensus guidelines, and recent advances in retrieval-augmented generation (RAG), explainable AI (XAI), and agentic clinical architectures, this document synthesizes what the literature establishes, where gaps remain, and how the proposed system contributes to trustworthy health informatics. The survey is conducted in the context of a research prototype using simulated patient cases and guideline-style evidence; it does not claim clinical validation or deployment.

1. Introduction

1.1 Research Context

Clinical decision support systems (CDSS) have long aimed to assist clinicians with evidence-based recommendations at the point of care. With the rapid adoption of machine learning and large language models (LLMs), CDSS capabilities have expanded substantially. However, a persistent gap separates predictive performance from clinical trustworthiness. Many systems still function as black boxes: they may produce accurate risk scores or recommendations, but they often fail to show how outputs were generated, whether they are supported by clinical evidence, or when they should not be trusted.

This literature survey supports the design of a Self-Auditing Clinical Decision Support Agent—a system that transforms the traditional pipeline:

Input → Model → Output

into:

Input → Prediction → Evidence → Reasoning → Verification → Output

The survey is structured to inform each architectural layer of the proposed system and to position the work within the broader movement toward trustworthy health informatics, including alignment with the Texas Health Informatics Alliance (THIA) 2026 theme: From Innovation to Impact: Implementing Trustworthy Health Informatics Across Texas.

1.2 Scope and Methodology

This survey focuses on peer-reviewed literature, consensus guidelines, and high-quality preprints published primarily between 2018 and 2026. Search themes included:

- Clinical decision support and evidence-based medicine
- Explainable and trustworthy AI in healthcare
- Clinical information retrieval and guideline grounding
- Retrieval-augmented generation (RAG) in biomedicine
- Self-verification, claim auditing, and reliability scoring
- Agentic and modular AI architectures for clinical workflows

Sources were identified through database searches (PubMed, IEEE Xplore, ACM Digital Library, arXiv), forward citation tracking, and systematic review reference lists. This is a narrative literature survey in progress; a formal systematic review protocol (e.g., PRISMA) may be applied in later phases.

1.3 Research Questions Guiding the Survey

No.	Research Question
RQ1	Why is prediction alone insufficient for trustworthy clinical AI?
RQ2	What methods exist for grounding clinical AI in evidence?
RQ3	How do current systems verify or audit their own reasoning?
RQ4	What architectural patterns support transparent, modular clinical AI?
RQ5	What evaluation frameworks exist for trustworthiness beyond accuracy?
RQ6	Where are the gaps that a self-auditing CDS architecture can address?

2. Background: Clinical Decision Support and the Trust Gap

2.1 Evolution of Clinical Decision Support

Clinical decision support has evolved from rule-based expert systems and encoded clinical practice guidelines (CPGs) to data-driven machine learning models and, more recently, LLM-based assistants. Early CDSS research emphasized knowledge representation—transforming guideline knowledge into computable rules and inference engines (Georg et al., 2004; Hashmi & Zrimec, 2008; Hussain & Abidi, 2007). These systems offered transparency through explicit rules but struggled with scalability, maintenance, and adaptation to patient-specific context.

Modern ML-based CDSS improved predictive accuracy across domains including cardiovascular risk, sepsis, and diagnostic imaging. Yet interpretability concerns intensified as models grew more complex. La Cava et al. (2023) note that ML models trained for clinical decision support are typically "either accurate or interpretable but not both," highlighting a fundamental tension that continues to shape the field.

2.2 The Black-Box Problem in Healthcare AI

The black-box problem is well documented. Eke and Shuib (2025), in a systematic literature review on explainability and transparency in AI healthcare systems, identify that clinicians require not only accurate

predictions but also understandable explanations of how outputs relate to patient data and clinical knowledge. Their review surfaces persistent open issues: lack of standardized explainability metrics, inconsistent evaluation of explanation quality, and limited integration of explanations into clinical workflows.

Similarly, Rosenbacke et al. (2024), cited in a 2026 perspective on translation-ready healthcare AI (Molecular Systems Biology), found that while some XAI studies report increased clinician trust, others show no significant effect or even diminished trust when explanations are incoherent or overly complex. This nuance challenges the assumption that "more explanation" automatically improves adoption.

2.3 Clinician Trust as a Multidimensional Requirement

A 2025 JMIR systematic review on trust in AI-based CDSS among healthcare workers (27 studies) identifies eight themes critical to trust:

System transparency — clear, interpretable AI behavior

Training and familiarity — user education and knowledge sharing

System usability — workflow integration

Clinical reliability — consistency and accuracy

Credibility and validation — performance across diverse contexts

Ethical consideration — fairness, liability, standards adherence

Human-centric design — patient-centered approaches

Customization and control — preserving clinician autonomy

Barriers to trust include algorithmic opacity, insufficient training, and ethical challenges. This review establishes that trust is not a byproduct of accuracy—it is a system-level property requiring intentional design across technical, organizational, and human factors.

Relevance to this project: The proposed Self-Auditing Clinical Decision Support Agent directly targets the transparency, clinical reliability, and credibility themes by making evidence alignment and audit results explicit in every output.

3. Theme I: Interpretable Clinical Risk Prediction

3.1 Logistic Regression as a Baseline for Clinical Risk Models

Logistic regression remains a cornerstone of clinical risk prediction due to its interpretability, probabilistic output, and direct mapping of coefficients to odds ratios (Hosmer et al., 2013; ScholasticaHQ, 2024).

Established clinical scores such as CHADS₂ and Framingham risk models demonstrate the enduring value of transparent statistical models in evidence-based practice.

A 2024 case study on deploying interpretable AI in healthcare diagnostics compared logistic regression, Explainable Boosting Machines (EBM), and XGBoost with SHAP on the UCI Heart Disease dataset. While XGBoost achieved superior predictive performance, EBM offered a better balance of accuracy, transparency, and clinical usability. Logistic regression was noted as particularly suitable for epidemiological and tabular clinical risk tasks where coefficient interpretability supports clinical reasoning.

La Cava et al. (2023) extended this line of work through symbolic regression (FEAT), demonstrating that interpretable models can achieve accuracy comparable to black-box approaches across 25 clinical prediction tasks in MIMIC-III. Their findings support the design principle that baseline CDS prototypes should prioritize interpretable prediction before introducing opaque models.

3.2 Post-Hoc Explainability and Its Limitations

Post-hoc methods such as SHAP, LIME, and Grad-CAM dominate XAI literature in CDSS. A 2025 meta-analysis of 62 studies (2018–2025) on XAI in CDSS found that model-agnostic techniques and attention mechanisms are widely used in radiology, oncology, neurology, and critical care. Studies combining high predictive performance with strong explanation fidelity (≥ 0.85) reported clinician trust scores 12–18 percentage points higher than those without quantitative explanation evaluation.

However, the same meta-analysis identifies critical gaps: few studies evaluate explanation fidelity, clinician trust, or usability in real-world settings. Post-hoc explanations can be clinically incoherent, sensitive to noise, or misleadingly increase confidence in incorrect predictions (FUTURE-AI guideline; Vellido, 2020).

3.3 Synthesis for Risk Prediction Layer

Approach	Strengths	Limitations	Relevance to Project
Logistic Regression	Interpretable, fast, probabilistic, supports feature contribution analysis	May underperform compared to complex models	Selected baseline
Explainable Boosting Machines / GAMs	Balance of accuracy and interpretability	Less familiar to clinicians and may require additional explanation	Future upgrade
XGBoost + SHAP	High accuracy with local post-hoc explanations	Explanations are post-hoc and may not guarantee clinical reasoning fidelity	Comparison baseline
Deep Learning	Strong performance in imaging and sequential clinical data	Opaque, harder to audit, higher compute cost	Out of scope for Phase 1

Research gap: Most risk prediction literature focuses on model-level interpretability (coefficients, SHAP values) but does not connect predictions to external clinical evidence or verify whether downstream recommendations align with retrieved guidelines.

4. Theme II: Evidence Retrieval and Guideline Grounding

4.1 Clinical Information Retrieval

Clinical information retrieval (IR) supports clinicians in finding relevant evidence from guidelines, literature, and EHR corpora at the point of care. A comprehensive literature review on clinical IR (TMC Digital Commons) documents that TF-IDF and BM25 remain the most widely used vector-space retrieval methods, together appearing in more than 70 reviewed papers. These methods represent documents and queries as term-weighted vectors and rank results by similarity (typically cosine similarity).

TF-IDF weights terms by their frequency in a document relative to their rarity across the corpus, making it effective for identifying clinically salient terminology (Sparck Jones, 1972). BM25 extends TF-IDF with document length normalization and probabilistic tuning, and is frequently used in hybrid retrieval pipelines.

4.2 TF-IDF for Clinical Practice Guidelines

Direct evidence supports TF-IDF-based retrieval for guideline question answering. A 2025 study using Gensim TF-IDF and Latent Semantic Indexing (LSI) models to retrieve answers from international CPGs reported that the true answer appeared in the first six retrieved results in 98% of cases. The authors note that CPGs are reliable evidence-based sources—more than 90% of relevant clinical questions can be answered from guidelines—but their volume limits point-of-care accessibility without IR support.

A multi-classifier guideline sentence classification system (PMC, 2012) demonstrated that integrating TF-IDF search with sentence-level classifiers improved the probability of surfacing relevant guideline content compared to TF-IDF alone, though at increased runtime cost.

4.3 Retrieval-Augmented Generation in Healthcare

RAG has emerged as the dominant paradigm for grounding LLM outputs in external medical knowledge. A 2025 JAMIA systematic review and meta-analysis of 20 RAG studies in biomedicine found a pooled odds ratio of 1.35 (95% CI: 1.19–1.53) for RAG-improved performance over baseline LLMs, with statistically significant effect ($p = 0.001$). The authors recommend future work on system-level enhancements combining RAG with agents, deeper knowledge integration, and EHR-integrated deployment.

A 2026 narrative review on RAG in healthcare (Digital Medicine) describes the evolution from naive RAG to advanced and modular RAG paradigms. RAG addresses core LLM limitations—hallucination, outdated knowledge, and lack of citation traceability—by retrieving evidence before generation. The framework supports continuous knowledge base updates without full model retraining.

Recent systems demonstrate guideline-grounded RAG at scale. A 2025 system for querying UK NICE clinical guidelines achieved MRR of 0.814 and 99.1% recall within the top ten retrieved chunks using a hybrid BM25 + dense embedding architecture across 10,195 text chunks from 300 guidelines.

4.4 Limitations of Retrieval Alone

Retrieval improves factual grounding but does not guarantee that generated reasoning faithfully uses retrieved evidence. Liu et al. (2024, Bioinformatics) introduced retrieval-augmented LLMs with self-reflection for medical reasoning, showing that retrieval plus reflection outperforms retrieval alone—but also demonstrating that reflection quality varies by task.

Furthermore, VERIRAG (2025) argues that standard RAG systems are methodologically blind: they retrieve clinically relevant documents but cannot assess study quality, detect retractions, or weight evidence by methodological rigor. This is a critical gap for high-stakes clinical decision support.

4.5 Synthesis for Evidence Retrieval Layer

Method	Strengths	Limitations	Project Role
TF-IDF + Cosine Similarity	Simple, interpretable, fast, lightweight, useful for clinical guideline retrieval	Limited semantic understanding and weak paraphrase handling	Phase 1 baseline
BM25	Robust probabilistic ranking with document length normalization	Still primarily lexical and not fully semantic	Phase 2 upgrade
Dense Embeddings	Better semantic matching and paraphrase understanding	Less explainable, requires more compute, may be harder to audit	Phase 2 upgrade
Hybrid Retrieval + Reranking	Stronger recall and precision by combining lexical and semantic signals	More complex pipeline and harder to validate	Phase 2-3 upgrade

Research gap: Most RAG literature evaluates retrieval relevance and answer accuracy but provides limited frameworks for downstream verification that generated claims are actually supported by retrieved passages.

5. Theme III: Explainable and Evidence-Constrained Reasoning

5.1 From Explanation to Evidence-Constrained Generation

Traditional XAI provides explanations after a model produces output. Evidence-constrained reasoning inverts this priority: the system is designed so that claims must be grounded in retrieved evidence before presentation. This aligns with the FUTURE-AI principle of explainability, which requires that AI explanations be evaluated both quantitatively (computational fidelity) and qualitatively (impact on clinician confidence and performance).

The FUTURE-AI international consensus guideline (BMJ, 2024)—developed by 117 experts from 50 countries—defines six principles for trustworthy healthcare AI: fairness, universality, traceability, usability, robustness, and explainability. Its 30 best practices span the full AI lifecycle, from need definition through post-deployment monitoring. Traceability—maintaining records of data, training, and audit logs—is directly relevant to self-auditing architectures.

5.2 Structured Reasoning in Clinical AI

Structured reasoning approaches decompose clinical tasks into explicit steps rather than generating free-form text. MEDDxAgent (ACL, 2025) introduces a modular explainable differential diagnosis agent with an orchestrator, history-taking simulator, and specialized agents for knowledge retrieval and diagnostic strategy. Interactive evaluation showed >10% accuracy improvements over single-turn approaches, with critical explainability into the diagnostic reasoning process.

Multi-agent clinical frameworks further decompose responsibility. A 2025 multi-agent CDSS architecture for ICU settings uses specialized agents for lab analysis, vital signs interpretation, and contextual reasoning, with a central integration agent and a validation agent (arXiv, 2025). This mirrors clinical practice where physicians gather evidence from multiple sources before forming a unified assessment.

MAM (2025) decomposes multimodal medical diagnosis into role-specialized agents (General Practitioner, Specialist Team, Radiologist, Medical Assistant, Director), achieving 18–365% performance improvements over baseline models through modular collaboration.

5.3 The "No Evidence → No Claim" Principle

The proposed system's core reasoning constraint—no evidence, no claim—draws on established principles from evidence-based medicine and fact-checking in NLP. In clinical practice, recommendations should be traceable to guideline statements or peer-reviewed evidence. In AI systems, this translates to:

Generating claims only when supporting evidence is retrieved

Flagging claims that exceed retrieved evidence

Surfacing missing evidence for important clinical concerns

This principle is increasingly reflected in verification-focused RAG research (see Theme IV) but is not yet standard in CDS prototypes.

Research gap: Few clinical AI systems enforce evidence constraints at the architecture level; most rely on prompt engineering or post-hoc filtering without systematic audit.

6. Theme IV: Self-Auditing, Verification, and Reliability Scoring

6.1 The Emerging Field of AI Self-Verification

Self-verification refers to systems that evaluate their own outputs before presenting them to users. Wornow et al. (2023) demonstrated that a self-verification pipeline for clinical information extraction—using omission detection, evidence grounding, and pruning of unsupported elements—improved accuracy across multiple LLMs including GPT-4. Critically, the pipeline grounds each extracted element to a specific text span, enabling efficient human auditing.

This work establishes a foundational insight for the proposed project: there is an asymmetry between generation and verification—it is often easier for a model to check whether a claim is supported than to generate correct claims in the first place.

6.2 Claim-Level Verification in Biomedical RAG

Recent work moves from document-level to claim-level verification:

VERIRAG (2025) introduces healthcare claim verification via statistical audit in RAG. It addresses the limitation that RAG retrieves evidence but cannot judge methodological quality. VERIRAG uses: (i) an 11-point Veritable audit for methodological rigor; (ii) a Hard-to-Vary (HV) score weighting evidence by quality and diversity; and (iii) a Dynamic Acceptance Threshold calibrated to claim boldness. Across four corpora, VERIRAG achieved F1 scores of 0.53–0.65, representing 10–14 point improvements over baselines including Self-RAG, FLARE, and CIBER.

MedRAGChecker (2026) decomposes biomedical RAG answers into atomic claims and evaluates each using natural language inference (NLI) combined with biomedical knowledge-graph consistency signals. It produces answer-level diagnostics including faithfulness, under-evidence, contradiction, and safety-critical error rates. This framework directly parallels the proposed self-auditing layer's goals of claim-evidence alignment and unsupported claim detection.

Self-RAG (Asai et al., 2023) introduced retrieval, generation, and critique through self-reflection tokens, enabling models to retrieve on demand and critique their own outputs. While designed for general domains, it established the architectural pattern of embedding verification into the generation loop.

6.3 Rule-Based Verification for Transparency

Rule-based verification offers advantages in healthcare contexts where inspectability and auditability are paramount:

- Transparent logic that can be reviewed by clinicians and regulators
- Deterministic behavior for safety-critical checks
- No additional model inference cost
- Easy integration with reliability scoring

The proposed project's Phase 1 rule-based audit layer aligns with this literature, serving as a baseline before introducing learned verification (NLI, ensemble verifiers) in later phases.

6.4 Reliability Scoring Frameworks

Reliability scoring synthesizes multiple audit signals into a single trust indicator. Potential factors identified across the literature include:

Factor	Description	Literature Basis
Claim Support Rate	Measures the proportion of generated claims supported by retrieved evidence	MedRAGChecker-style claim verification
Evidence Relevance	Measures how relevant retrieved evidence is to the patient case or generated recommendation	RAG and clinical IR literature
Unsupported Claim Detection	Identifies claims that cannot be traced back to retrieved evidence	Self-verification and biomedical RAG verification research
Contradiction Detection	Flags claims that conflict with retrieved evidence or clinical constraints	Claim-level verification frameworks
Prediction-Recommendation Consistency	Checks whether the predicted risk level aligns with the suggested recommendation	Proposed project contribution
Safety Rule Violations	Detects unsafe outputs such as medication dosage instructions or unsupported diagnostic claims	Trustworthy AI and clinical safety frameworks
Explanation Fidelity	Evaluates whether the explanation accurately reflects the model output and retrieved evidence	XAI and FUTURE-AI principles

FUTURE-AI recommends that explainability methods be evaluated quantitatively before qualitative clinician assessment, and that limitations of explanations (incoherence, noise sensitivity, overconfidence) be explicitly identified.

6.5 Synthesis for Self-Auditing Layer

The literature establishes that:

- Retrieval alone is insufficient — evidence must be verified at the claim level
- Self-verification improves reliability — asymmetric verify-vs-generate is exploitable

Audit should precede presentation — not replace clinician judgment, but surface weak reasoning

Scoring enables governance — reliability metrics support accountability and regulatory alignment

Research gap: Existing verification frameworks (VERIRAG, MedRAGChecker) focus on biomedical QA and literature claims. Few integrate verification into a full CDS pipeline that combines structured patient data, risk prediction, guideline retrieval, and clinical recommendation generation with end-to-end reliability scoring.

7. Theme V: Agentic and Modular Clinical AI Architectures

7.1 From Monolithic Models to Agentic Pipelines

Agentic AI refers to systems that decompose complex objectives into specialized, coordinated modules or agents capable of autonomous reasoning within defined boundaries. In healthcare, this paradigm addresses limitations of monolithic models by separating responsibilities across prediction, retrieval, reasoning, verification, and output generation.

A 2025 agentic AI framework for end-to-end medical data inference orchestrates clinical workflows through domain-specialized agents handling data ingestion, anonymization, model training, inference, and explanation. The framework emphasizes modality-aware preprocessing and interpretable outputs, reducing dependence on expert intervention.

Industry analysis (2026) positions agentic AI on a deployment timeline where clinical decision support requires robust governance and is categorized as medium-term (2027–2028) relative to lower-risk administrative automation. This underscores that agentic CDS architectures must be designed with governance from inception, not as an afterthought.

7.2 Multi-Agent CDSS with Governance

The 2025 multi-agent CDSS paper for ICU settings explicitly integrates ethical AI governance principles—autonomy, fairness, and accountability—into its architecture. Specialized agents analyze labs, vitals, and clinical context; a prediction agent generates structured outputs; and a validation agent cross-checks results. This architecture closely parallels the proposed pipeline and provides empirical support for modular agent design in clinical environments.

7.3 Architectural Comparison

Architecture	Prediction	Retrieval	Reasoning	Verification	Output Type
Traditional ML CDSS	Yes	No	No	No	Risk score or classification
LLM Chatbot	No	No	Yes	No	Free-text response
Basic RAG System	No	Yes	Yes	No	Grounded answer
Self-RAG	No	Yes	Yes	Partial	Critiqued generated response
Proposed Self-Auditing CDS Agent	Yes	Yes	Yes	Yes	Transparent clinical decision-support output with evidence and audit trace

Research gap: Agentic clinical architectures are emerging rapidly, but few combine interpretable prediction (logistic regression) with lightweight retrieval (TF-IDF) and rule-based self-audit in a single, demonstrable pipeline designed for research transparency and conference presentation.

8. Theme VI: Evaluation, Governance, and Translation Readiness

8.1 Beyond Accuracy: Trustworthiness Evaluation

The 2026 perspective on designing academic research for translation readiness argues that healthcare AI must address trustworthiness proactively during study design, not post hoc. Key dimensions include:

Explainability — stakeholder-specific, not one-size-fits-all

Uncertainty quantification — communicating confidence and limitations

Evaluation — measuring human-AI team performance, not isolated algorithm metrics

The XAI meta-analysis (2025) recommends longitudinal clinical validation, participatory system design, and uniform interpretability metrics. The JMIR trust review (2025) calls for diverse demographics, cross-cultural perspectives, and profession-specific trust factors.

8.2 Proposed Evaluation Metrics for This Project

Based on the literature, the following metrics are appropriate for Phase 4 evaluation with simulated cases:

Metric	Description	Literature Basis
Retrieval Relevance	Measures whether retrieved guideline passages are relevant to the patient case using metrics such as Precision@k or MRR	Clinical IR and RAG evaluation literature
Evidence Support Rate	Measures the percentage of generated claims supported by retrieved evidence	Claim-level biomedical RAG verification
Unsupported Claim Rate	Measures the percentage of claims flagged by the audit layer as unsupported	Self-verification and evidence-grounding literature
Audit Reliability	Measures agreement between automated audit verdicts and manual human review	Verification and auditability frameworks
Prediction-Recommendation Consistency	Evaluates whether the recommendation is logically aligned with the predicted risk level	Proposed project metric
Reliability Score Distribution	Analyzes the spread and behavior of generated reliability scores across simulated cases	Trustworthy AI evaluation
Safety Violation Rate	Measures how often the system generates unsafe or out-of-scope outputs	Clinical AI safety and governance principles

8.3 Governance and Safe Positioning

The literature consistently warns against overclaiming in healthcare AI. Safe positioning for this research prototype includes:

Describing the system as a decision-support tool, not a diagnostic engine

Using simulated patient cases for demonstration

Stating that outputs require clinician review

Not claiming clinical validation, deployment, or outcome improvement unless empirically demonstrated

This aligns with FUTURE-AI lifecycle guidance and the EU AI Act's emphasis on documentation, traceability, and intended-use specifications.

9. Research Gaps and Contribution of This Work

9.1 Identified Gaps

Gap ID	Gap	Description	Literature Evidence
G1	Prediction-Reasoning Disconnect	Risk models and recommendation systems are often separated, with limited verification between predicted risk and generated reasoning	Clinical prediction and XAI literature

Gap ID	Gap	Description	Literature Evidence
G2	Retrieval Without Verification	RAG improves grounding but does not guarantee that generated claims are supported by retrieved passages	RAG and biomedical verification literature
G3	Post-Hoc Explainability	Many systems rely on explanations after prediction rather than enforcing evidence constraints during reasoning	XAI and trustworthy AI guidelines
G4	Missing Reliability Scoring	Current systems often lack structured reliability scores for clinical AI outputs	Trust and governance literature
G5	Monolithic Architectures	Single-model systems are harder to inspect, debug, and govern	Agentic and modular AI research
G6	Limited End-to-End CDS Evaluation	Few systems evaluate the full pipeline from patient input to prediction, retrieval, reasoning, audit, and output	Clinical AI evaluation literature

9.2 Contribution of the Self-Auditing Clinical Decision Support Agent

This project addresses the identified gaps through an integrated architecture that:

Combines interpretable prediction with evidence retrieval — logistic regression risk scores inform evidence queries and reasoning context

Enforces evidence-constrained reasoning — the "no evidence → no claim" rule is architectural, not optional

Introduces a dedicated self-auditing verification layer — rule-based claim-evidence matching with a path to NLI-based verification

Produces reliability scores — composite trust indicators based on audit results

Generates transparent clinical output — risk score, concerns, recommendations, evidence citations, audit trace, and safety disclaimer in a single structured output

Uses a modular agentic pipeline — separable components for prediction, retrieval, reasoning, auditing, and scoring that support independent evaluation and iteration

9.3 Positioning Relative to Prior Work

Prior Work	Primary Focus	This Project's Differentiation
VERIRAG	Healthcare claim verification in scientific literature RAG	Extends verification into a full CDS-style pipeline with patient data, risk prediction, guideline retrieval, and audit scoring
MedRAGChecker	Claim-level verification for biomedical RAG answers	Applies claim-evidence alignment principles to clinical decision-support outputs
MEDDxAgent	Modular agent framework for explainable differential diagnosis	Focuses more broadly on trustworthy clinical decision support with self-auditing verification
FUTURE-AI	International guideline for trustworthy healthcare AI	Operationalizes traceability, explainability, and auditability in a prototype architecture
TF-IDF CPG Retrieval	Guideline-based clinical question answering	Integrates retrieval into a multi-stage audited clinical AI pipeline
Self-Verification Research	Verification for clinical information extraction	Extends verification from extraction tasks to recommendation-level clinical reasoning

10. Literature-Informed Architecture Mapping

The following table maps surveyed literature themes to the proposed system architecture:

Architecture Layer	Literature-Informed Basis
Patient Data Input	Structured EHR, vitals, labs, and standard CDS input formats
Clinical Feature Processing	Normalization and interpretable model preparation - Hosmer et al. (2013); FEAT, La Cava et al. (2023)
Risk Prediction (Logistic Regression)	Interpretable baseline - ScholasticaHQ (2024); IJSATE (2024)
Evidence Retrieval (TF-IDF + Cosine Similarity)	Clinical practice guideline IR - Gensim TF-IDF/LSI (2025); Clinical IR review; NICE RAG (2025)
Structured Reasoning (Evidence-Constrained)	Modular agents - MEDDxAgent (2025); Multi-agent CDSS (2025)
Self-Auditing Verification (Rule-Based -> NLI)	Claim verification - VERIRAG (2025); MedRAGChecker (2026); Self-Verification (2023)
Reliability Scoring	Trust metrics - FUTURE-AI (2024); JMIR trust review (2025)
Transparent Clinical Output	Governance-aware presentation - FUTURE-AI; THIA 2026 theme

11. Future Literature Directions

As this survey continues, the following areas require deeper review:

- Hybrid retrieval methods — BM25 + dense embeddings, reranking, citation tracking
- NLI-based claim verification — adapting MedRAGChecker-style ensembles for CDS outputs
- Human factors studies — how reliability scores and audit traces affect clinician trust and decision-making
- Regulatory frameworks — EU AI Act, FDA SaMD guidance, and HAARF for agentic clinical AI
- Evaluation benchmarks — simulated patient case corpora with annotated evidence-claim pairs
- Fairness and bias — ensuring risk prediction and retrieval do not disproportionately affect demographic subgroups (FUTURE-AI fairness principle)

12. Conclusion

This literature survey establishes a clear research foundation for the Self-Auditing Clinical Decision Support Agent. The surveyed literature converges on several key findings:

- Prediction is necessary but insufficient. Clinician trust requires transparency, evidence grounding, and verifiable reasoning—not just accuracy.
- Retrieval grounds AI in evidence but does not verify reasoning. RAG reduces hallucination but introduces new risks when systems generate claims not supported by retrieved passages.
- Self-verification is an emerging and promising paradigm. Claim-level auditing, evidence alignment, and reliability scoring represent the frontier of trustworthy clinical AI.
- Modular agentic architectures improve auditability. Separating prediction, retrieval, reasoning, and verification enables independent evaluation and governance.
- Significant gaps remain. No existing system fully integrates interpretable risk prediction, lightweight evidence retrieval, evidence-constrained reasoning, and self-auditing verification into a single transparent CDS pipeline designed for research demonstration.

The proposed project is positioned to address these gaps through a honest, research-grade prototype that prioritizes system trustworthiness over model complexity. As the literature survey continues and the prototype is implemented, empirical evaluation on simulated patient cases will test whether self-auditing verification measurably reduces unsupported clinical reasoning—a question that prediction-only systems cannot answer.

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